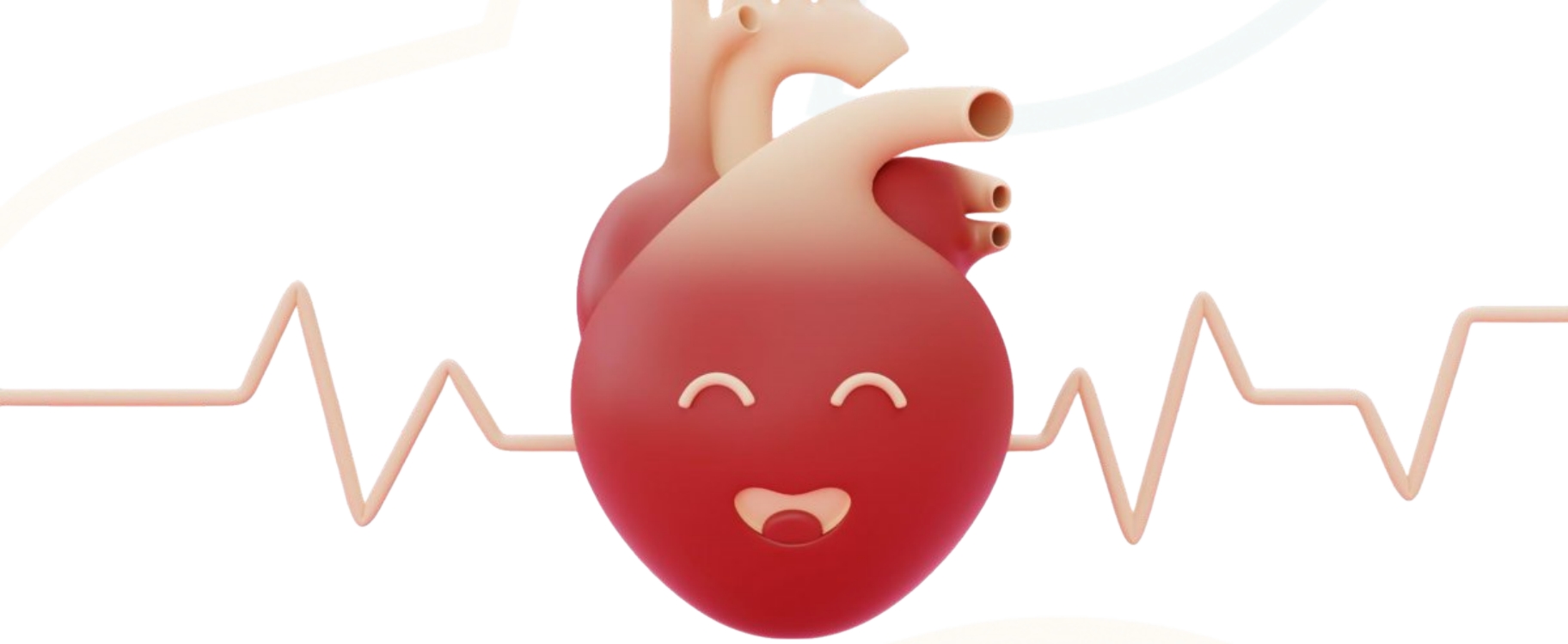
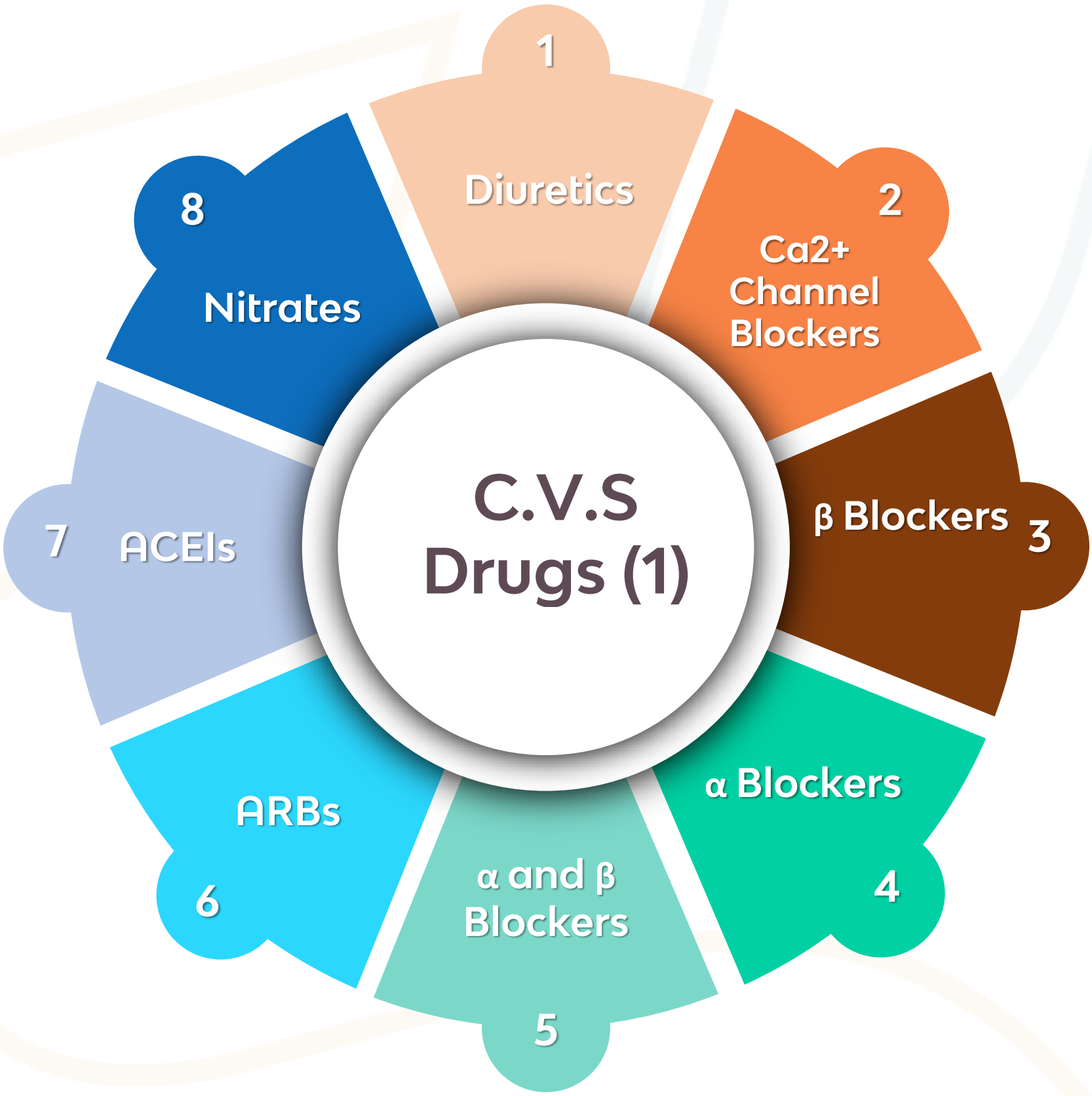
Cardiovascular Drugs (Part 2)



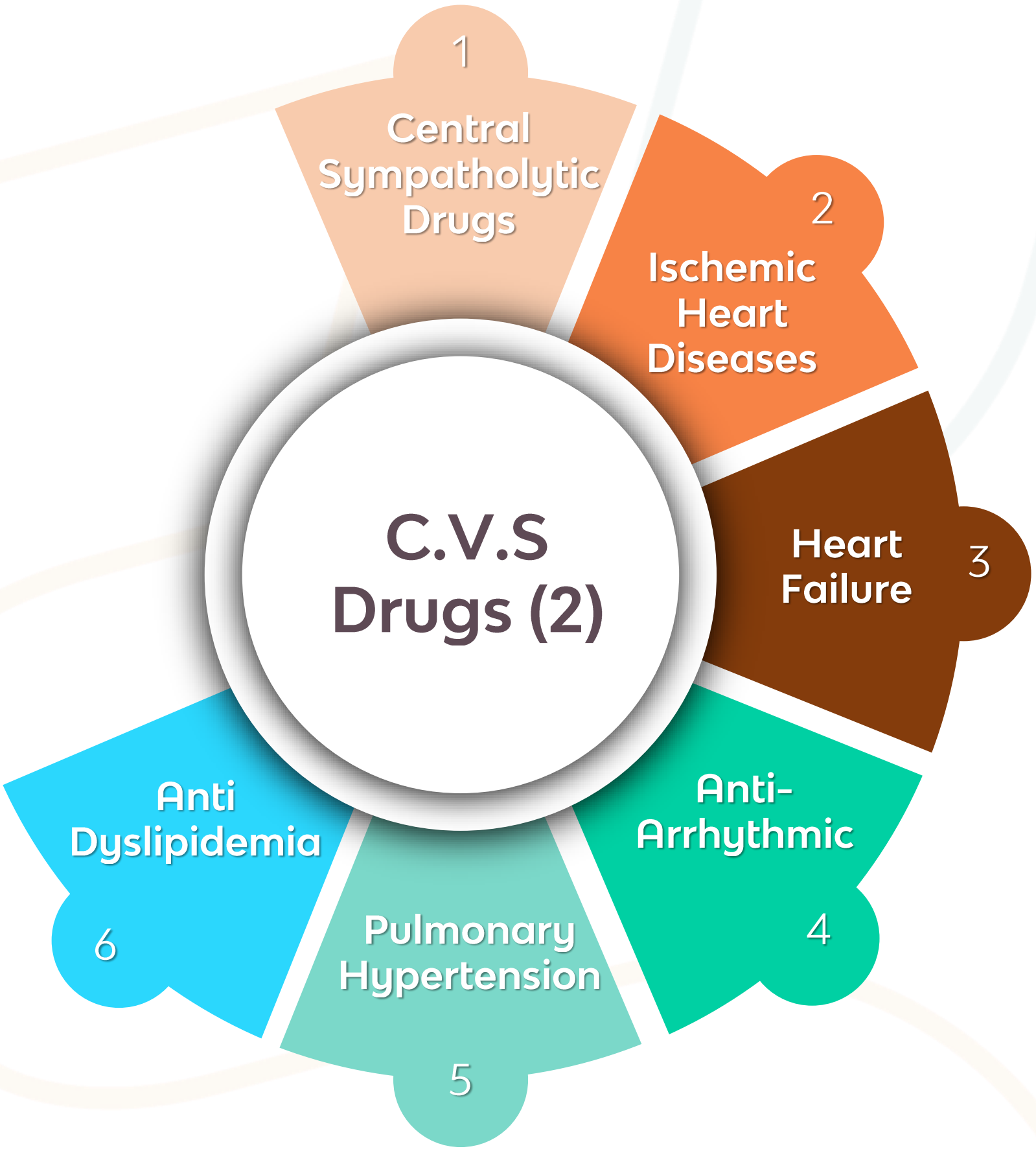
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United Pharmacy

Talent Management Department

CVS drugs classification



CVS drugs classification



Today we will learn about

01 Central Sympatholytic Drugs

02 Ischemic Heart Diseases

03 Heart Failure Drugs

04 Anti-Arrhythmic

05 Anti Dyslipidemia

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Central Sympatholytic Drug

Act on the brainstem to reduce sympathetic outflow → ↓ heart rate & blood pressure
Examples: Clonidine, Methyldopa

Use: Resistant hypertension, pregnancy (methyldopa)

Note: Can cause sedation, dry mouth, and rebound hypertension if stopped suddenly.

Drugs for Ischemic Heart Diseases

Improve oxygen supply to heart or reduce oxygen
Used in angina (stable, unstable), post-MI, and to prevent complications

1

Central Sympatholytic Drugs

- **ALDOMET®|** Methyldopa

2

Drugs for Ischemic Heart Diseases

- **VASTAREL ®|** Trimetazidine

3

Drugs for Heart Failure

Heart Load Regulator "ARB + Neprilysin Inhibitor"

- **ENTRESTO®|** Sacubitril & Valsartan

Antiarrhythmic Drugs

Goal is to restore or maintain normal heart rhythm
Each class works by modifying how ions flow through cardiac cells, affecting conduction and/or refractoriness.

Most of these classes are unavailable but for your knowledge.

Class I – Sodium Channel Blockers (Na^+ blockers):
↓ conduction velocity

Class II – Beta Blockers: ✅
↓ sympathetic stimulation → ↓ SA node rate, AV conduction, and

Class III – Potassium Channel Blockers: ✅
↑ action potential duration & refractory period

Class IV – Calcium Channel Blockers (Non-DHP):
↓ AV node conduction & automaticity

Other (Unclassified): ✅
Adenosine: ↑ K^+ efflux, ↓ Ca^{2+} influx at AV node → transient AV block (used in PSVT)
Digoxin: Increases vagal tone → slows AV conduction (also ↑ contractility)

Most of these drugs won't be in the regular community pharmacies yet to be found in the hospital pharmacy, so read it for your knowledge.

4

Anti-Arrhythmic Agents

- **Class I**
  - **Class IA**
    - *Procainamide*
    - *Quinidine*
    - *Disopyramide*
  - **Class IB**
    - *Mexiletine*
  - **Class IC**
    - *Flecainide*
    - *Propafenone*
- **Class II**
  - **See β Blockers** (in day 10).
- **Class III**
  - **CORDARONE®** | *Amiodarone*
  - **VASODARONE®** | *Dronedarone*
  - **BETACOR®** | *Sotalol*
- **Class IV**
  - **See Calcium Channel Blockers (Diltiazem & Verapamil)** (in day 10).

CORDARON® *is Available in the pharmacy !*

Anti Dyslipidemia

1. Statins (HMG-CoA Reductase Inhibitors): ✅
Inhibit **HMG-CoA reductase**, the key enzyme in cholesterol synthesis↓ intracellular cholesterol → ↑ LDL receptors on hepatocytes → ↑ LDL clearance from blood
Effect: ↓ LDL (major), ↓ TG (modest), ↑ HDL (mild)
Note: First-line; monitor liver enzymes, watch for myopathy (SE).

2. Fibrates: ✅
Activate PPAR-α receptor → ↑ lipoprotein lipase activity↑ breakdown of VLDL → ↓ triglycerides, ↑ HDL synthesis.
Effect: ↓ TG, ↑ HDL
Note: Watch for myopathy (especially with statins).

Anti Dyslipidemia

3. Cholesterol Absorption Inhibitor ✅

- Inhibits **NPC1L1 protein in the intestinal wall** → ↓ cholesterol absorption from the gut
- Liver senses ↓ cholesterol → ↑ LDL receptor expression → ↑ LDL uptake from blood
- **Effect**: ↓ LDL (moderate)

4. PCSK9 Inhibitors (Injectables) ✅

- Inhibit PCSK9, a protein that degrades LDL receptors More LDL receptors remain on hepatocytes → ↑ LDL clearance
- Effect: ↓↓ LDL (very potent)

Note: Used in **familial hypercholesterolemia** or statin intolerance.

5. Omega-3 Fatty Acids (Fish Oil) ✅

- Inhibit VLDL synthesis and enhance TG clearance
- Effect: ↓ TG
- Used in severe hypertriglyceridemia; may cause fishy taste, ↑ bleeding risk at high doses

5 Therapy for Dyslipidemia

- **Statins**
  - **ATORLIP , LORVAST** ®| Atorvastatin
  - **ZOCOR** ® | Simvastatin
  - **CRESTOR** ® | Rosuvastatin
  - **LESCOL** ® | Fluvastatin
  - **LIVAZO** ® | Pitavastatin
- **Fibrates**
  - **LOPID** ® | Gemfibrozil
  - **LIPANTHYL** ® | Fenofibrate
- **Cholesterol Absorption Inhibitor**
  - **EZETROL** ®, **EZECHOL** ®| Ezetimibe
- **PCSK-9* Inhibitors**
  - **REPATHA** ® | Evolocumab
  - **PRALUENT** ® | Alirocumab
- **JP OMEGA 3** ®| Omega-3
- **Combinations**
  - **INEGY** ® | Ezetimibe + Simvastatin
  - **ATOZET** ®| Atorvastatin + Ezetimibe

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط